

Pathway ID	Intervention	Expected benefit	Contextual response	Possible environmental consequence	Classification	Mitigation	Evidence basis	Confidence
P1	Technically repairable hardware, hinges, handles or fittings	Longer service life because short-life components can be replaced without discarding the full window	Physical access is blocked by finishes, reveals, wall interface or installation detailing	Full-window replacement may occur even though the failed part is technically repairable	Implementation gap	Design for component access, maintenance clearance and documented repair sequence	Supplier and architect interviews, component map, VELFAC case uncertainty	Medium
P2	Replaceable components and spare parts	Product-life extension through future repair and component replacement	Compatible spare parts become unavailable or difficult to source after the window system becomes older	The complete window may be replaced earlier than technically necessary	Implementation gap	Long-term spare-part guarantee, compatibility requirements and supplier commitment	Supplier interview, lifecycle assumptions and codebook category spare parts	Medium
P3	Replaceable visible or functional components	Easier maintenance, adjustment and targeted upgrading	Repair opportunity becomes an opportunity for cosmetic or performance upgrading before functional failure	Components or whole windows may be replaced more frequently than needed	Potential behavioural rebound	Replacement criteria, repair-first hierarchy and user guidance	Interview evidence on client preference for newer and more modern systems	Low to medium
P4	Highly modular or high-performance window system	Easier disassembly, better performance and possible component replacement	Added connectors, materials, mechanisms or technical complexity are introduced to enable modularity or higher performance	Higher initial production burden may reduce or offset part of the expected lifecycle benefit	Lifecycle trade-off	Minimise added materials and verify benefit through lifecycle scenario comparison	Architect interview, lifecycle scenario comparison and literature coding on material complexity	Medium
P5	Long-life frame retained for extended service life	Reduced need for full-window replacement and lower whole-system material demand	Future glazing or performance upgrades may be restricted by the retained frame, interface or product architecture	The building may keep a technically outdated window longer or require disruptive replacement later	Technological lock-in	Design for future glazing replacement, upgrade compatibility and performance verification	VELFAC case, component map and lifecycle assumptions on frame and glazing life	Medium
P6	Component repair and maintenance strategy	Lower material demand compared with full replacement	Multiple service visits, specialist labour or long-distance transport are needed over the study period	Expected material savings may be partly reduced by transport, labour and repeated intervention burdens	Burden shifting	Group repair visits, support local service infrastructure and include transport assumptions in scenario comparison	Transparent assumptions table and lifecycle scenario comparison	Low to medium
P7	Reuse of removed full windows or major components	Avoided new production through direct reuse or refurbishment	Project-specific dimensions, certification uncertainty, ageing components and transport make reuse difficult	Reuse potential may not be realised or may create additional adaptation and transport burdens	Implementation gap or burden shifting	Dimensional standardisation, local reuse networks, documentation and certification routes	Supplier and architect interviews, reuse exclusion from Scenario C	Medium